

Churn Prediction

Comprehensive Analysis & Insights

Report Generated:	December 23, 2025
Platform:	Insightify Professional Analytics
Engine Version:	3.0 (Enhanced AI Analytics)
Report Status:	Confidential

■ Executive Summary

Report Overview:

This comprehensive data analysis report presents key insights and actionable metrics from the provided dataset. Our analysis covers **10,000** records across **14** columns, utilizing advanced statistical methods and visualization techniques.

Dataset Composition:

- Numeric Variables: **11** columns
- Categorical Variables: **3** columns
- Temporal Variables: **0** columns
- Data Completeness: **0** missing values detected
- Data Uniqueness: **0** duplicate rows identified
- Memory Footprint: **2.64 MB**

■ Key Insights & Findings

Our analysis has uncovered the following significant patterns:

1. Age shows significant right-skewed distribution (skewness: 1.01)
2. High variability detected in Exited (std: 0.40, mean: 0.20)
3. Exited shows significant right-skewed distribution (skewness: 1.47)
4. Geography is highly concentrated: 'France' represents 50.1% of data
5. Gender is highly concentrated: 'Male' represents 54.6% of data

■ Numeric Analysis

Column	Mean	Median	Std Dev	Min	Max	IQR
RowNumber	5000.50	5000.50	2886.90	1.00	10000.00	4999.50
CustomerId	15690940.57	15690738.00	71936.19	15565701.00	15815690.00	124705.50
CreditScore	650.53	652.00	96.65	350.00	850.00	134.00
Age	38.92	37.00	10.49	18.00	92.00	12.00
Tenure	5.01	5.00	2.89	0.00	10.00	4.00
Balance	76485.89	97198.54	62397.41	0.00	250898.09	127644.24
NumOfProducts	1.53	1.00	0.58	1.00	4.00	1.00
HasCrCard	0.71	1.00	0.46	0.00	1.00	1.00
IsActiveMember	0.52	1.00	0.50	0.00	1.00	1.00
EstimatedSalary	100090.24	100193.91	57510.49	11.58	199992.48	98386.14
Exited	0.20	0.00	0.40	0.00	1.00	0.00

■ Categorical Analysis

Category: Surname

Unique Values: 2932 | Most Frequent: 'Smith' (0.3% of data)

Value	Count	Percentage
Smith	32	12.0%
Scott	29	10.9%
Martin	29	10.9%
Walker	28	10.5%
Brown	26	9.7%
Genovese	25	9.4%
Yeh	25	9.4%
Shih	25	9.4%
Wright	24	9.0%
Maclean	24	9.0%

Category: Geography

Unique Values: 3 | Most Frequent: 'France' (50.1% of data)

Value	Count	Percentage
France	5,014	50.1%
Germany	2,509	25.1%
Spain	2,477	24.8%

Category: Gender

Unique Values: 2 | Most Frequent: 'Male' (54.6% of data)

Value	Count	Percentage
Male	5,457	54.6%
Female	4,543	45.4%

■ Correlation Analysis

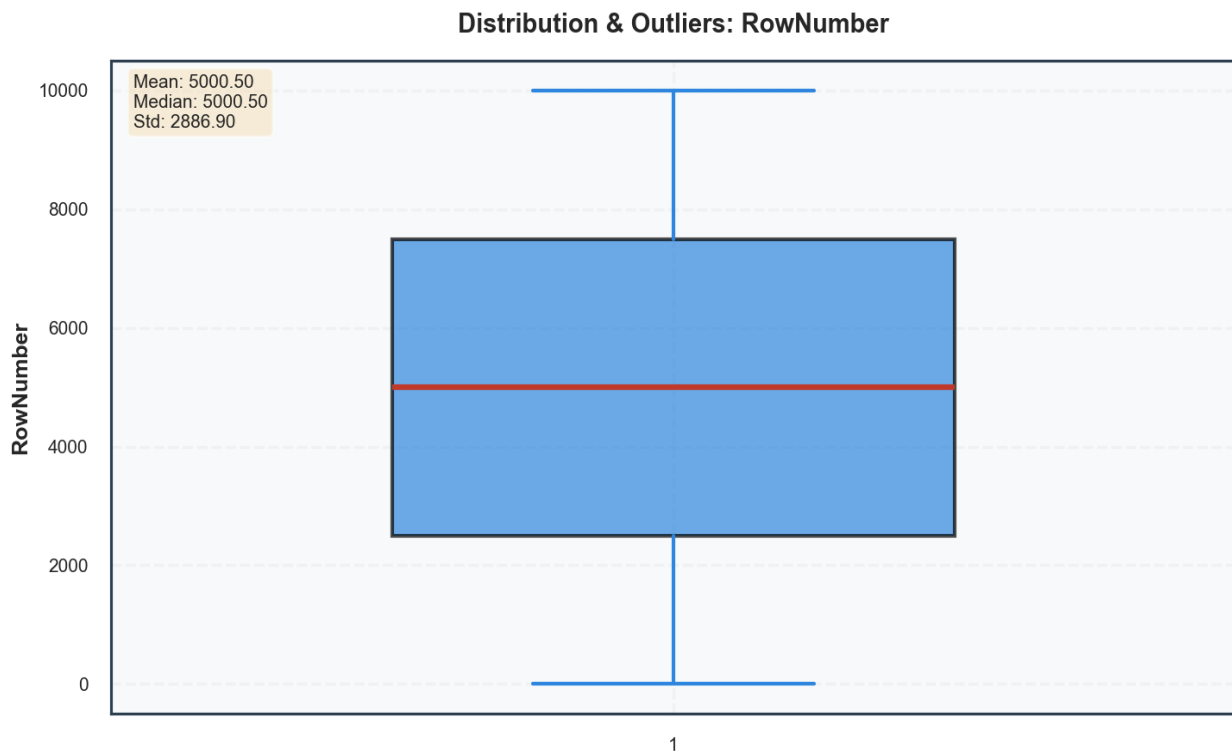
Identified 1 significant relationships between variables:

Correlation coefficients range from -1 (perfect negative) to +1 (perfect positive).

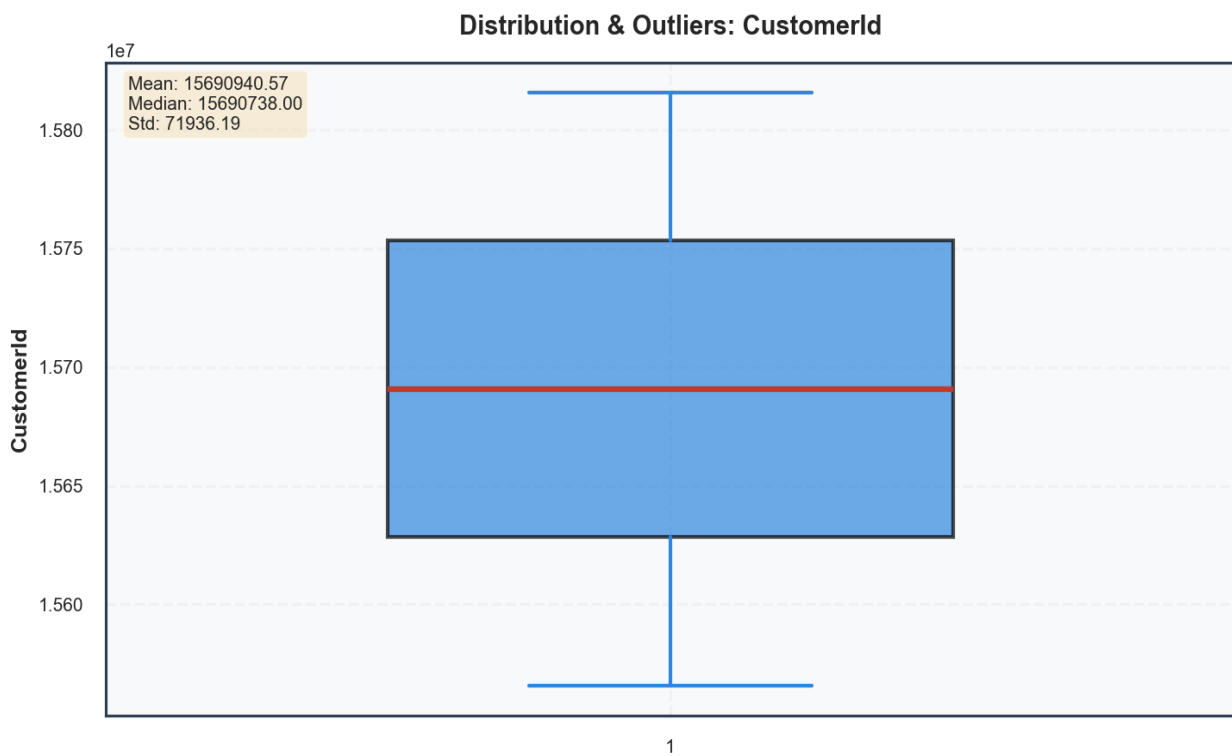
Variables Pair	Coefficient	Strength	Direction
Balance vs NumOfProducts	-0.3042	Weak	Negative

■ Data Visualizations

Boxplot Rownumber



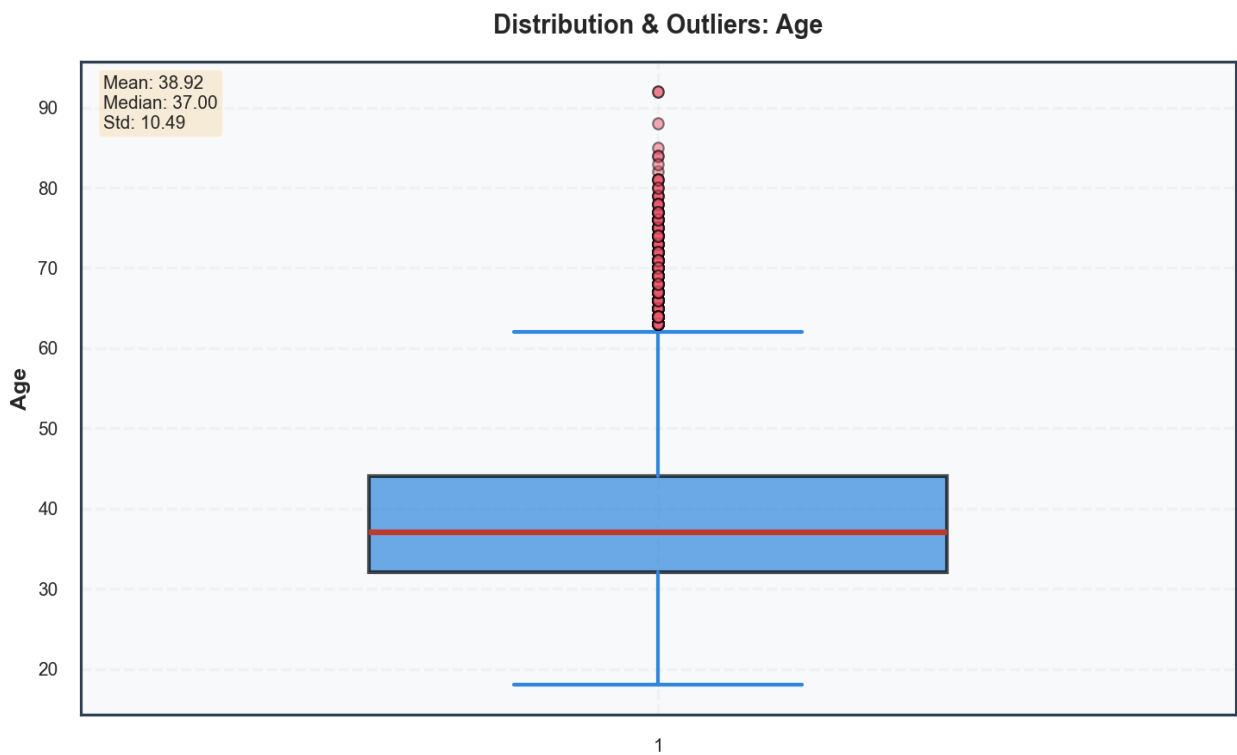
Boxplot CustomerId



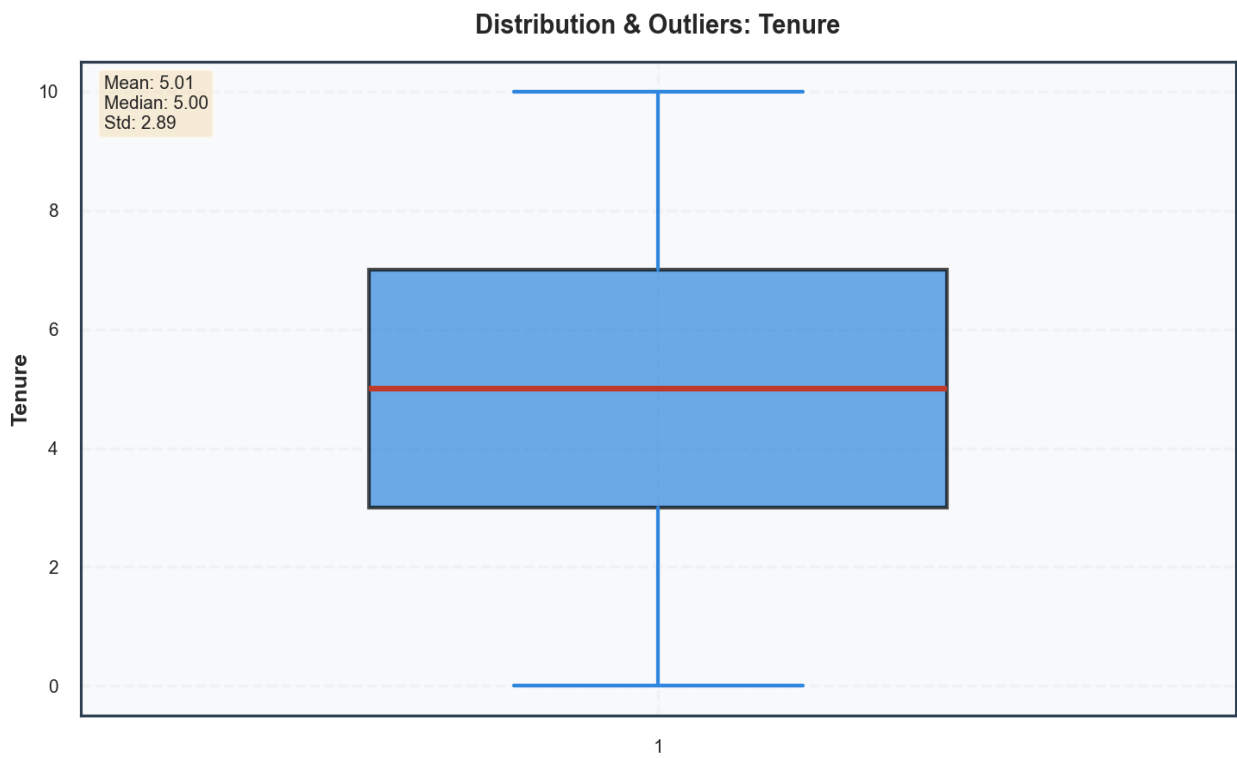
Boxplot Creditscore



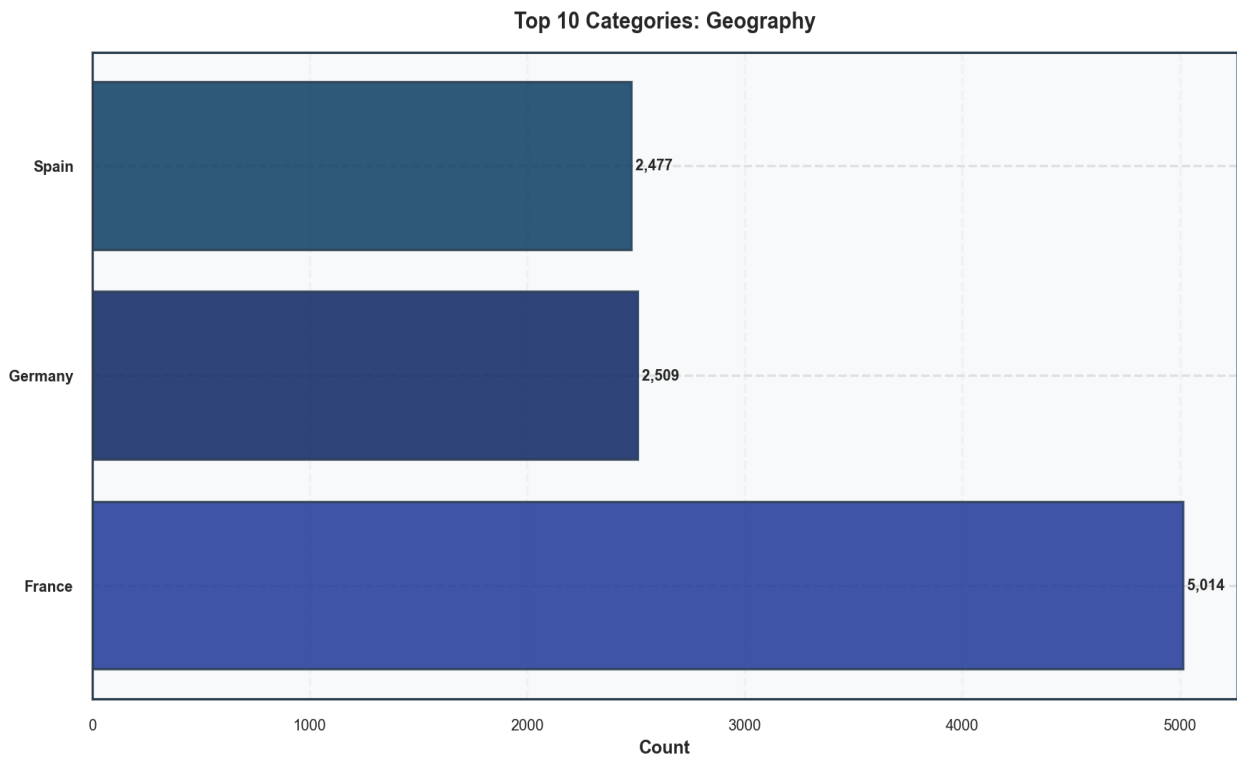
Boxplot Age



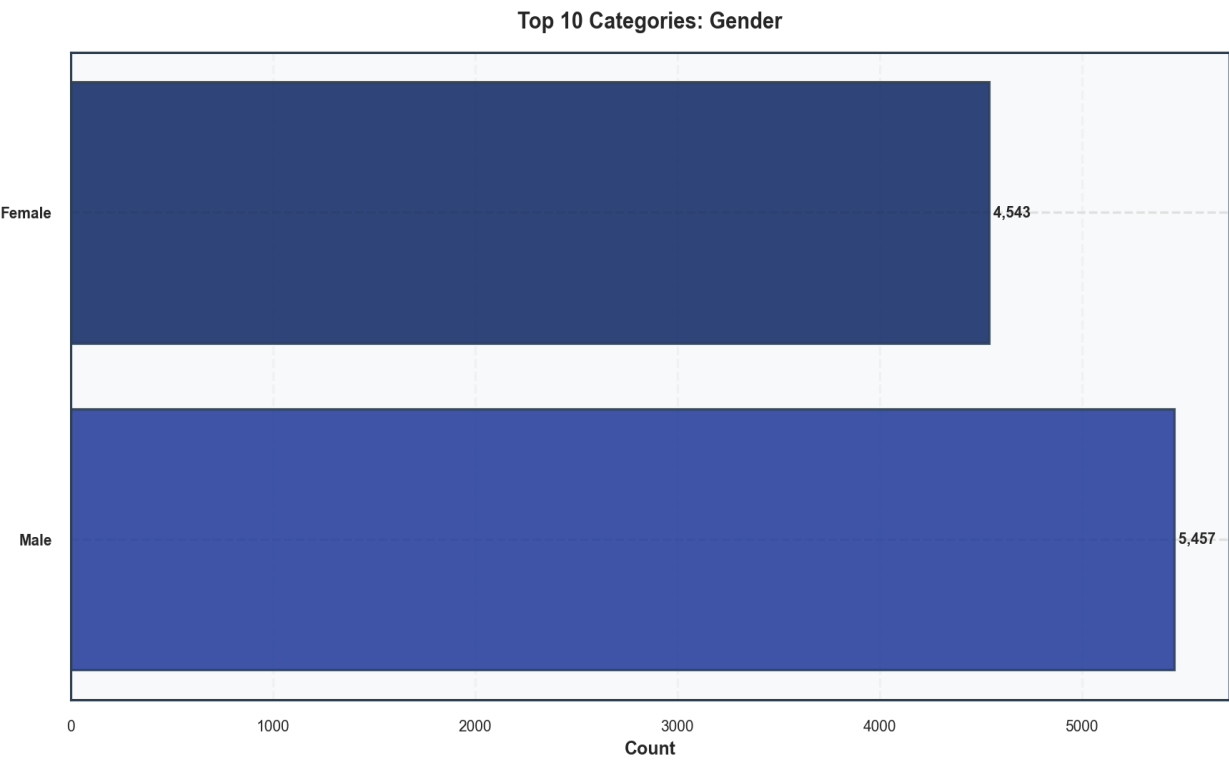
Boxplot Tenure



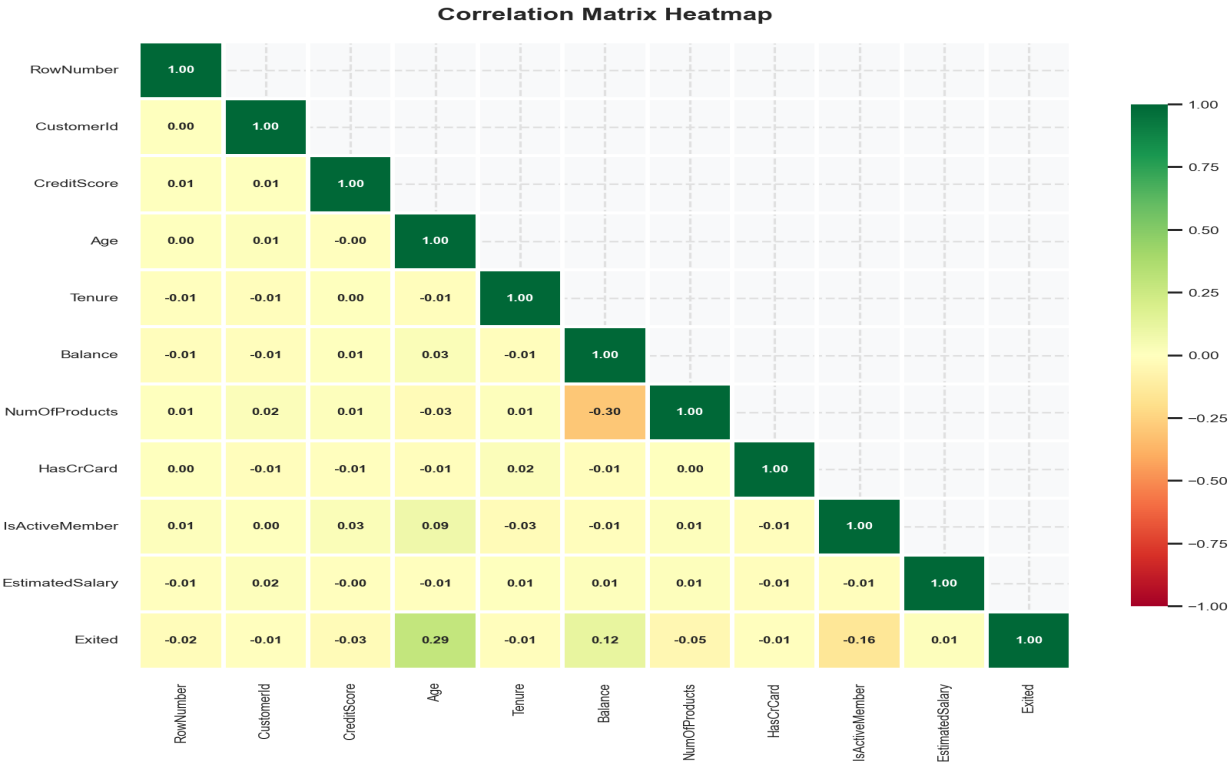
Bar Geography



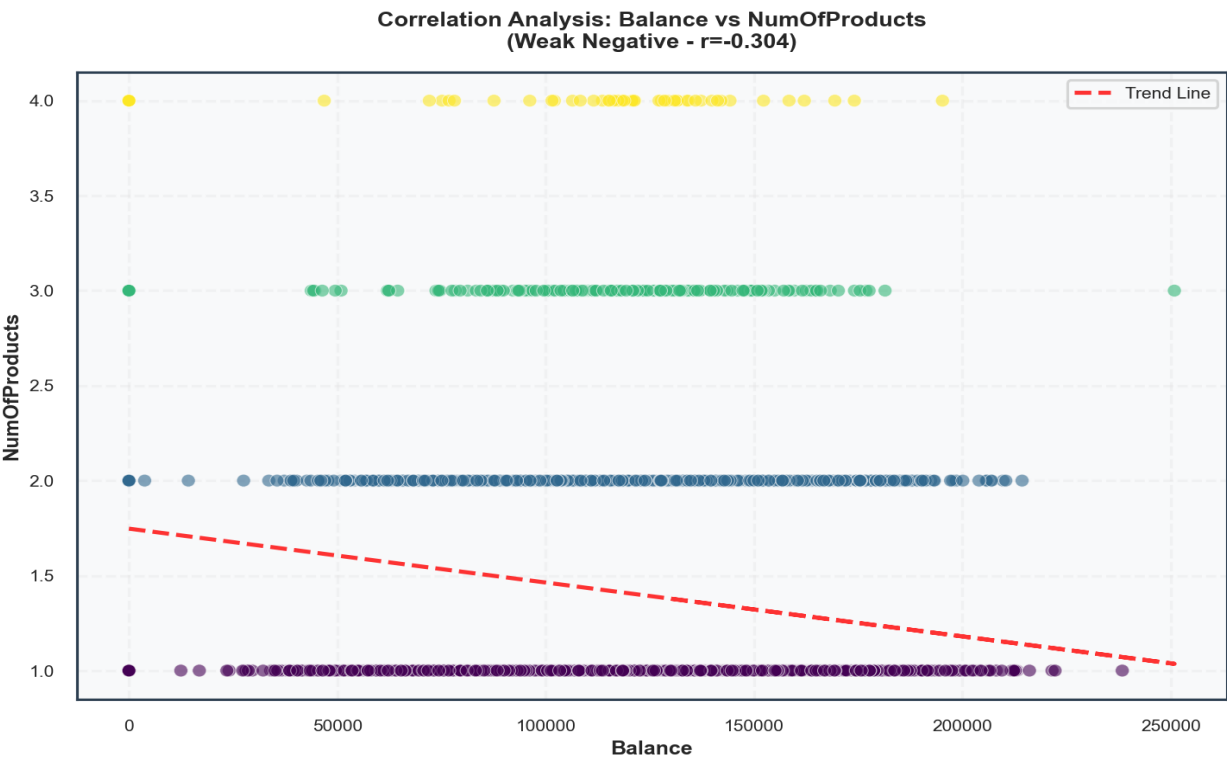
Bar Gender



Correlation Heatmap



Scatter Balance Vs Numofproducts



■ Conclusions & Recommendations

Key Findings:

1. Data Quality Assessment:

The dataset has undergone comprehensive quality checks. Review the Executive Summary for critical insights regarding data completeness, duplicate records, and overall integrity. Address any missing values or duplicates identified before proceeding with predictive modeling.

2. Statistical Distribution Analysis:

Our numeric analysis reveals the central tendencies, variability, and distribution patterns across all quantitative variables. Box plots highlight potential outliers that may require further investigation or treatment depending on your analytical objectives.

3. Correlation & Relationship Discovery:

The correlation analysis has identified statistically significant relationships between variables. Strong positive or negative correlations ($|r| > 0.7$) indicate potential predictive relationships that could be leveraged in machine learning models or business intelligence applications.

Actionable Recommendations:

✓ **Investigate High-Correlation Pairs:** Variables with strong correlations warrant deeper analysis to determine causality versus mere association. Consider domain expertise when interpreting these relationships.

✓ **Address Data Quality Issues:** Implement data cleaning protocols for missing values and duplicates. Consider imputation strategies or removal based on the nature and extent of missing data.

✓ **Leverage Visualizations for Stakeholder Communication:** The generated charts provide clear, professional representations of your data suitable for executive presentations and decision-making forums.

✓ **Monitor Categorical Distributions:** Highly concentrated categories or imbalanced classes may require special handling in predictive models (e.g., SMOTE for oversampling, stratified sampling).

✓ **Temporal Pattern Exploration:** If time-series trends are present, consider seasonal decomposition and forecasting techniques to predict future patterns.

Report Generated: 2025-12-23 02:19:30

This report contains proprietary and confidential business information. Distribution should be limited to authorized personnel only. For questions about this analysis, please contact your data analytics team.